

Foundation (Jalapeno)

Q1. Solve the inequality $|x - 2| < 3$.

- (A) $-1 < x < 5$
 (B) $-3 < x < 1$
 (C) $0 < x < 4$
 (D) $1 < x < 3$
 (E) $2 < x < 6$

(B) $0 \leq d \leq 20$ (C) $8 \leq d \leq 12$ (D) $3 \leq d \leq 17$ (E) $10 \leq d \leq 20$ Q2. The distance between a basketball player and the hoop is given by $|d - 15|$. If the player is within 3 feet of the hoop, what is the range of possible distances?

- (A) $12 \leq d \leq 18$
 (B) $10 \leq d \leq 20$
 (C) $14 \leq d \leq 16$
 (D) $8 \leq d \leq 22$
 (E) $6 \leq d \leq 24$

Q6. Solve the inequality $|3x - 2| < 1$.

(A)

 $\frac{1}{3} < x < 1$ $\frac{2}{3} < x < \frac{4}{3}$ $\frac{1}{3} < x < \frac{4}{3}$

(A)

 $\frac{1}{3} < x < \frac{4}{3}$ $\frac{1}{3} < x < \frac{4}{3}$ (A) $0 < x < \frac{2}{3}$ $\frac{1}{3} < x < \frac{4}{3}$

(A)

 $\frac{1}{3} < x < 2$ Q3. Solve the inequality $|2x + 1| \geq 5$.

- (A) $x \leq -3$ or $x \geq 2$
 (B) $x \leq 1$ or $x \geq 3$
 (C) $x \leq 0$ or $x \geq 4$
 (D) $x \leq -2$ or $x \geq 1$
 (E) $x \leq -4$ or $x \geq 3$

Q7. The absolute value of a golfer's score is given by $|s - 70|$.

If the golfer's score is within 5 strokes of par, what is the range of possible scores?

(A) $65 \leq s \leq 75$ (B) $60 \leq s \leq 80$ (C) $70 \leq s \leq 80$ (D) $65 \leq s \leq 85$ (E) $60 \leq s \leq 90$ Q4. Graph the solution set of $|x| > 2$ on a number line.

- (A) $(-\infty, -2) \cup (2, \infty)$
 (B) $(-\infty, 2] \cup [2, \infty)$
 (C) $[-2, 2]$
 (D) $(-\infty, -1] \cup [1, \infty)$
 (E) $[-1, 1]$

Q8. Solve the inequality $|x + 4| > 2$.(A) $x < -6$ or $x > -2$ (B) $x < -2$ or $x > 2$ (C) $x \leq -6$ or $x \geq -2$ (D) $x \leq -4$ or $x \geq 0$ (E) $x \leq 0$ or $x \geq 4$ Q5. A tennis player must be within 5 meters of the net. If the player's distance from the net is given by $|d - 10|$, what is the range of possible distances?(A) $5 \leq d \leq 15$

Open Ended Questions (FRQ)

Q9. A swimmer's time is given by $|t - 50|$. If the swimmer's time is within 5 seconds of the record, what is the range of possible times? Show your work and provide the solution in interval notation.Q10. Solve the inequality $|2x - 5| \leq 3$ and graph the solution set on a number line. Provide the solution in interval notation.

Chef's Tips

- Provide clear instructions to students
- Emphasize the importance of showing work
- Allow students to use calculators for checking solutions